

AI-Based Universal JEE Question Generation System

System Design Document & Architecture

Objective: Automatically generate unlimited mathematically-valid JEE/NEET question variants from a single expert-authored question image.

1. Problem Statement

Background

Competitive examinations such as JEE Advanced, JEE Mains, and NEET require students to solve large volumes of conceptually diverse questions. Manual question creation is expensive, slow, and difficult to scale for modern EdTech platforms.

Problem	Description
Manual Authoring	Experts spend 30–60 minutes creating one question
Low Scalability	Large-scale question generation is difficult manually
Constraint Violations	Random number replacement breaks equations
Diagram Consistency	Generated diagrams often become visually inconsistent

2. System Overview

The architecture is designed as a two-phase AI pipeline. Phase 1 performs semantic understanding of the input question image, while Phase 2 performs controlled image regeneration using validated mathematical constraints.

1. Question Upload

Original question image input

2. GPT-4 Vision

Semantic JSON extraction

3. Constraint Engine

SymPy-based validation

4. Qwen2-VL

Layout-aware edit planning

5. Rendering Engine

Diagram + equation rendering

6. Final Output

Variant question images

3. Semantic JSON Generation

GPT-4 Vision converts the uploaded question image into a structured semantic JSON representation.

```
{ "subject": "Physics", "chapter": "Projectile Motion", "variables": [ { "name": "theta",  
  "range": [30, 80], "sampled_value": 45 } ], "constraints": [ "0 < theta < 90" ], "formula":  
  "R = u^2 * sin(2*theta) / g" }
```

4. Constraint Validation Engine

Constraint	Purpose
Pythagoras	Ensures triangle validity
Snell's Law	Maintains optics consistency
Positive Resistance	Prevents invalid circuit values
Angle Bounds	Maintains physically valid angles
Energy Conservation	Ensures mechanics correctness

5. Controlled Image Regeneration

Phase 2 generates the final variant question image while preserving the original diagram structure, educational intent, and visual consistency.

Stage	Description
Qwen2-VL	Generates structured image edit instructions
Region Mapping	Maps variables to exact image locations
Diagram Rendering	Uses Matplotlib/SVG for clean diagrams
Equation Rendering	Uses LaTeX/mathtext for equations
Pillow Compositor	Creates final PNG image

6. Technology Stack

Technology	Purpose
GPT-4 Vision	Semantic extraction
SymPy	Constraint validation
Qwen2-VL	Layout understanding
Matplotlib	Diagram rendering
LaTeX/mathtext	Equation rendering
Pillow	Image compositing
FastAPI	Backend orchestration
CUDA / A100	GPU acceleration

7. Final Unified Architecture

The final architecture integrates multimodal reasoning, symbolic mathematics, layout-aware editing, and deterministic rendering into a unified AI pipeline.

Component	Technology	Role
Semantic Extraction	GPT-4.1 Vision	Image → JSON
Constraint Engine	SymPy	Validation
Layout Understanding	Qwen2-VL	Region planning
Diagram Rendering	Matplotlib/SVG	Figure generation
Equation Rendering	LaTeX/mathtext	Readable equations
Image Compositing	Pillow	Final image generation

8. Conclusion

This architecture demonstrates a scalable AI-native pipeline for automated educational content generation. By combining multimodal reasoning, symbolic mathematics, and deterministic rendering, the system enables generation of unlimited, mathematically-valid JEE/NEET question variants while preserving educational quality and diagram consistency.